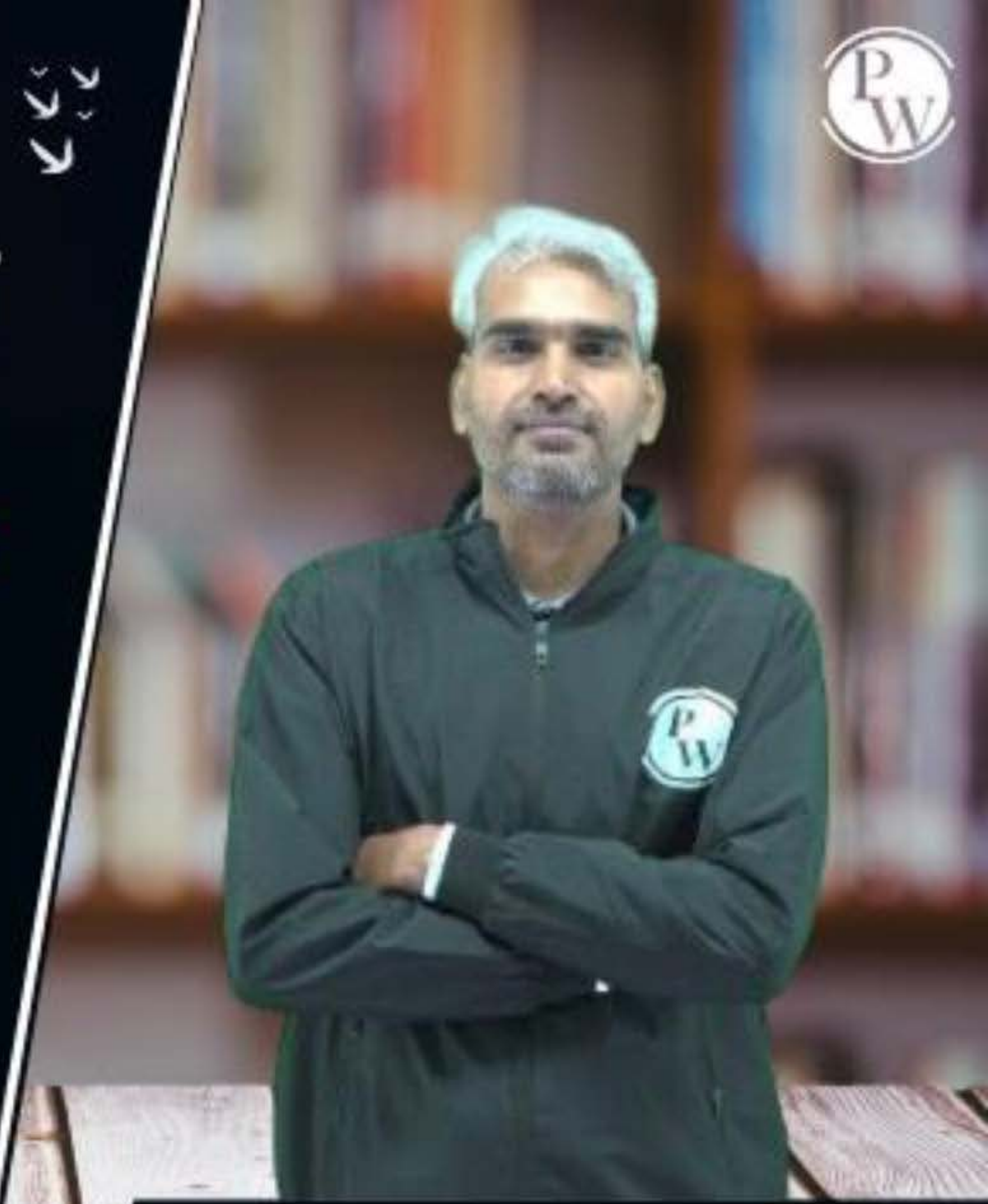
Computer Science & IT

Discrete Mathematics

Graph Theory

Lecture No. 03



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Recap of Previous Lecture

Topic Different types of graphs

Slide

Topics to be Covered

Topic GATE PYQ

Topic Degree sequence

Topic Graphic

Slide

Q.

Consider the set $S = \{1, w, w^2\}$, where w and w^2 are cube roots of unity. If * denotes the multiplication operation, the structure $\{S, *\}$ forms

MCQ [2010: 1 Mark]

(A) a group

(B) a ring

(C) an integral domain

(D) a field

Q.

Let G be a group with 15 elements. Let L be a subgroup of G. It is known that $L \neq G$ and that the size of L is at least 4. The size of L is 5.

NAT [2014 (Set-3): 1 Mark]

O(L) divides O(G) ✓

$\therefore$ Possible order = $\underbrace{1, 3}_{\times}$, 5, $\cancel{15}$

$O(L) \geq 4$

$L \neq G$

Q.

There are two elements x, y in a group (G,*)such that every element in the group can be written as a product of some number of x's and y's in some order. It is known that

$$\boxed{x*x = y*y = x*y*x*y = y*x*y*x = e}$$

Where e is the identity element. The maximum number of elements in such a group is 4.

2 / 4

NAT [2014 (Set-3): 2 Marks]

$\{x, y, e, \underset{y*x}{x*y}\}$

$(x*x*y*y) = (x*y*x*y)$

$\underbrace{x^{-1}*(x}_{e}*x*y*y) = \underbrace{x^{-1}*(x}_{e}*y*x*y)$

$(x*\underbrace{y*y)*y^{-1}}_{e} = (y*x*\underbrace{y)*y^{-1}}_{e}$

$\Rightarrow \boxed{x*y = y*x}$

Q.

Let G be a finite group on 84 elements. The size of a largest possible $\underbrace{\text{proper subgroup of G}}$ is _____.

NAT [2018: 1 Mark]

$\therefore O(\text{Sub-group}) \neq 84$

$\therefore$ Largest possible order $= \underline{\underline{42}}$

largest integer that divides 84

Q.

Let G be an arbitrary group.
Consider the following relations on G:
R_1: $\forall a, b \in G$, aR_1b if and only if $\exists g \in G$ such that $a = g^{-1}bg$
R_2: $\forall a, b \in G$, aR_2b if and only if $a = b^{-1}$
Which of the above is/are equivalence relation/relations?

MCQ [2019: 1 Mark]

(A) R_1 and R_2

(B) R_1 only

(C) R_2 only

(D) Neither R_1 nor R_2

└→ Reflexive $(a, a) \in R_1 \quad \forall a \in G$

$a \, R_1 \, a$ iff $g \in G$ s.t. $a = g^{-1} \cdot a \cdot g$

if G is a group. the 'e' $\in G$
s.t. $a = e^{-1} a \cdot e$
$\therefore$ Reflexive

$\underline{\underline{R_1}}$: Symmetric Rel^n.

if $(a, b) \in R_1$ then $(b, a) \in R_1$

$\underline{\underline{\text{i.e.}}}$ if there exist $a\ g \in G$ $\underline{\underline{\text{s.t}}}$ $a = g^{-1} * b * g$

$g * a = g * (g^{-1} * b * g)$

$g * a = b * g$

$g * a * g^{-1} = b * g * g^{-1}$

$$\boxed{(g^{-1})^{-1} a g^{-1} = b}$$

$\underline{\text{then}}$ there exist some λ $\underline{\underline{\text{s.t.}}}$ $b = \lambda^{-1} a \lambda$

$\underline{\text{yes}}$ and $\lambda = g^{-1}$

$\underline{\text{And}}$ if G is a group

$\underline{\text{then}}$ if $g \in G$ then $g^{-1} \in G$

$\underline{\underline{R_1}}$: $\underline{\text{Transitivity}}$:

if $(a,b) \in R_1$ & $(b,c) \in R_1$ then $(a,c) \in R_1$

$\underline{\text{let}}$ $g_1 \in G$ $\qquad$ $g_2 \in G$

s.t. $a = (g_1)^{-1} b\, g_1$ — (1) $\qquad$ s.t. $b = (g_2)^{-1} c\, g_2$ — (2)

Put this value of b in eqn (1)

$a = (g_1)^{-1}(g_2)^{-1} c\, g_2 g_1 \Rightarrow a = (g_2 g_1)^{-1} c\, (g_2 g_1)$

if $g_1, g_2 \in G$

$\underline{\text{then}}$ $g_2 g_1 \in G$ {closed}

$(g_2 g_1)^{-1} \in G$ {group}

$a = (g_2 g_1)^{-1} c\, (g_2 g_1)$

Q.

Let G be an arbitrary group.
Consider the following relations on G:
R_1: $\forall a, b \in G$, aR_1b if and only if $\exists g \in G$ such that $a = g^{-1}bg$
R_2: $\forall a, b \in G$, aR_2b if and only if $a = b^{-1}$
Which of the above is/are equivalence relation/relations?

MCQ [2019: 1 Mark]

(A) R_1 and R_2

(B) R_1 only

(C) R_2 only

(D) Neither R_1 nor R_2

$\underline{R_1}$: is an Equivalence Rel^n.

Check w.r.t. R_2

i.e. $(a, a) \in R_2$ iff $a = a^{-1}$

In an arbitrary group G,
Every element need not be inverse of itself.
$\therefore R_2$ need not be reflexive
$\therefore$ Need not be Equivalence Rel^n.

Q.

Let G be a group of 35 elements. Then the largest possible size of a subgroup of G $\underbrace{\text{other than G itself}}$ is ___7___. **NAT [2020: 1 Mark]**

$O(G) = 35$

Possible

$\therefore O(H) = 1, 5, 7, 35$

Q.

Let G be a group of order 6, and H be a subgroup of G such that $1 < |H| < 6$. Which one of the following options is correct?

Need not be

$o(H) = 1, \cancel{2}, 3, 6$

MCQ [2021: 2 Marks]

A. Both G and H are always cyclic.

cyclic

B. G is always cyclic, but H may not be cyclic.

Both are prime No.

of H is definitely cyclic

C. G may not be cyclic, but H is always cyclic.

D. Both G and H may not be cyclic.

Q. Which of the following statements is/are TRUE for a group G?

MSQ [2022: 1 Mark]

(A) If for all $x, y \in G$, $(xy)^2 = x^2y^2$, then G is commutative.

(B) If for all $x \in G$, $x^2 = 1$, then G is commutative. Here, 1 is the identity element of G.

(C) If the order of G is 2, then G is commutative.

(D) If G is commutative, then a subgroup of G need not be commutative.

True Statement: Sub-group of Commutative group is always Commutative.

$$(x*y)^2 = x^2 * y^2$$

$$x*y*x*y = x*x * y*y$$

$$\underbrace{x^{-1}*(x}_{e}*y*x*\underbrace{y)*y^{-1}}_{e} = \underbrace{x^{-1}(x}_{e}*x*y*\underbrace{y)*y^{-1}}_{e}$$

$$\boxed{y*x = x*y}$$

$Z = \{-\infty \ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$ w.r.t. Addition is Commutative

But $inv(1) = -1$ i.e $(1)^{-1} \neq 1$

---

if $x^2 = e$

i.e. if every element is inverse of itself then the group is Commutative.

$$(x * y) = (x * y)^{-1} = y^{-1} * x^{-1}$$

$$(y * x)$$

Q.

A pennant is a sequence of numbers, each number being 1 or 2. An n-pennant is a sequence of numbers with sum equal to n. For example, (1,1,2) is a 4-pennant. The set of all possible 1-pennants is {(1)}, the set of all possible 2-pennants is {(2), (1,1)}and the set of all 3-pennants is {(2,1), (1,1,1), (1,2)}.
Note that the pennant (1,2) is not the same as the pennant (2,1).
The number of 10-pennants is______. **NAT [2014 (Set-1): 2 Marks]**

Sequence can be of size

(5) $(2, 2, 2, 2, 2)$ only '1'

or (6) $(1, 1, 2, 2, 2, 2) \Rightarrow {}^{6}C_{4} * {}^{2}C_{2}$ such Pennants
$\hookrightarrow 15$

or (7) $(1, 1, 1, 1, 2, 2, 2) = {}^{7}C_{3} * {}^{4}C_{4}$ such Pennants
$\hookrightarrow 35$

or (8) $(1, 1, 1, 1, 1, 1, 2, 2) = {}^{8}C_{2} * {}^{6}C_{6}$ such Pennants
$\hookrightarrow 28$

or (9) $(1, 1, 1, 1, 1, 1, 1, 1, 2) = {}^{9}C_{1}$ such Pennants
$\hookrightarrow 9$

or (10) $(1, 1, 1, 1, 1, 1, 1, 1, 1, 1)$ only '1'

Total "89" 10-pennants

H.W. Q:- Let $A = \{1, 2, 3, 4, 5\}$

And let $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (3,2), (4,5)\}$ is a Partial order on set A.

How many total order relation are possible on set A such that relation 'R' is included in each total order relation

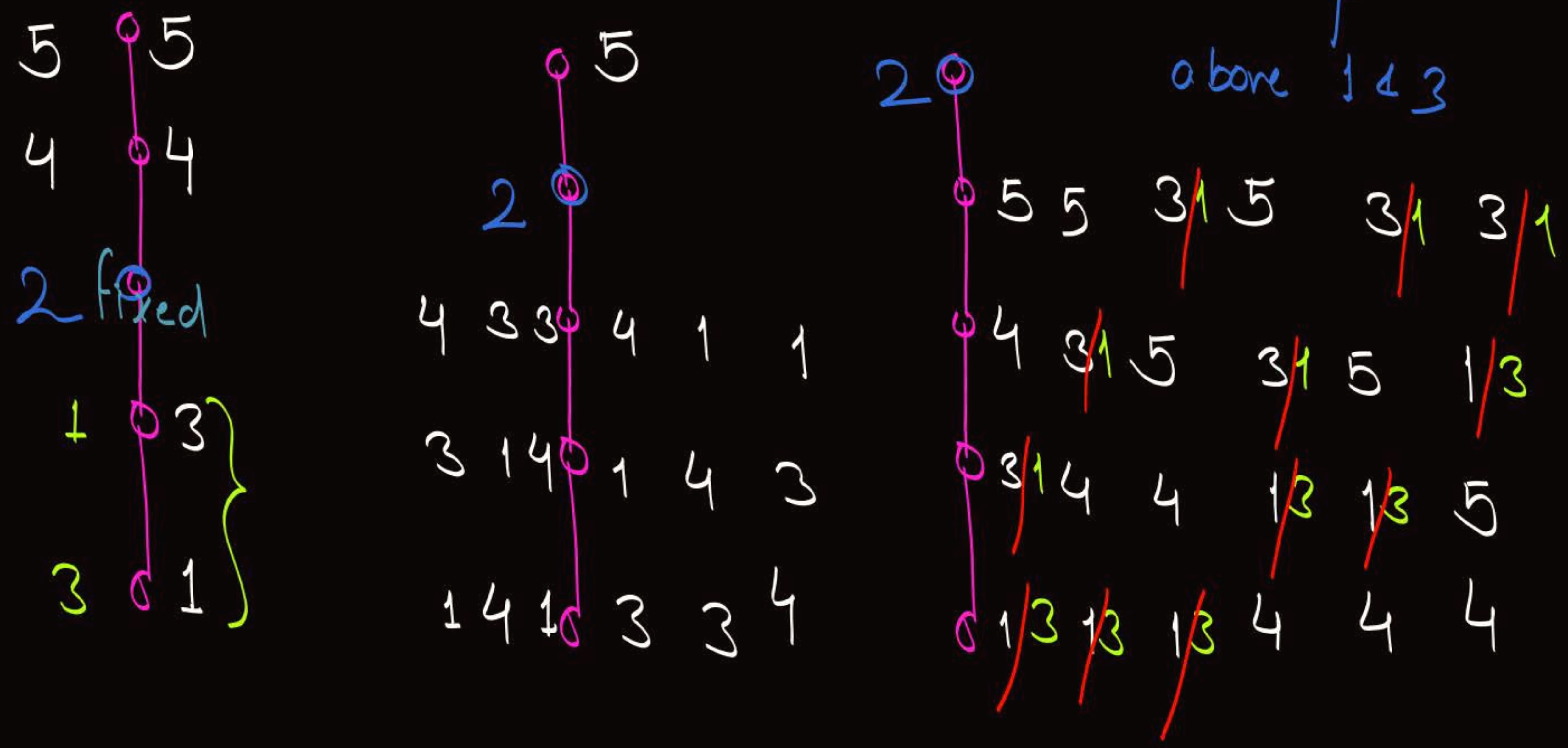
H.W. Q :-

Let $A = \{1, 2, 3, 4, 5\}$

And let $R = \{\underbrace{(1,1), (2,2), (3,3), (4,4), (5,5)}_{\text{No problem}}, \underbrace{(1,2), (3,2)}_{2}, \underbrace{(4,5)}_{5}\}$

2 above 1 & 3

5 above 4

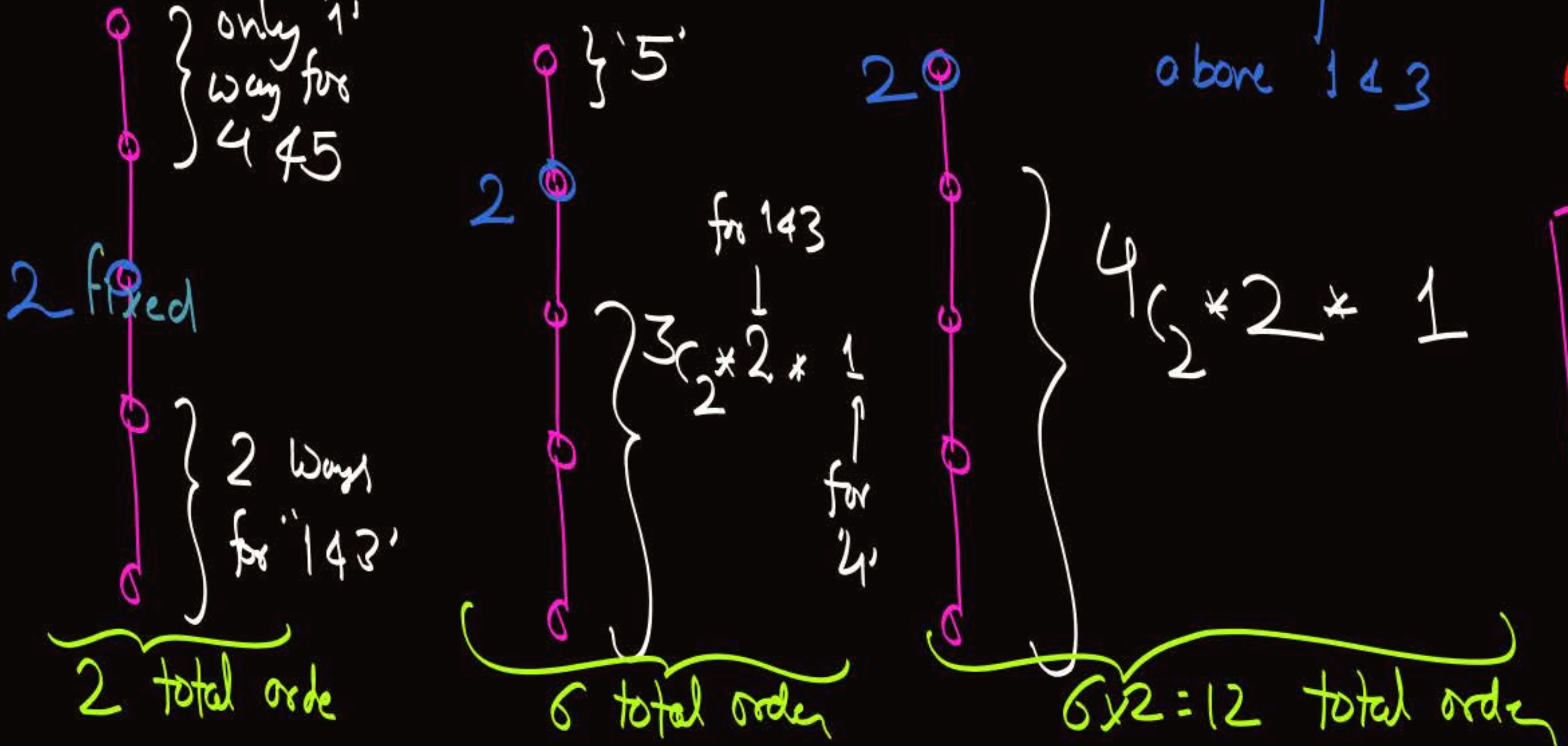


H.W. Q:- Let $A = \{1, 2, 3, 4, 5\}$

And let $R = \{\underbrace{(1,1), (2,2), (3,3), (4,4), (5,5)}_{\text{No problem}}, \underbrace{(1,2), (3,2)}_{2}, \underbrace{(4,5)}_{5}\}$

2 above 1 & 3

5 above 4



Total no. of total orders containing Partial order R is $= 2 + 6 + 12$

$= 20$

2
3
1

Only one choice of order for 1, 2 & 3

Q:- Let $A = \{1, 2, 3, 4, 5\}$

And let $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (1,3), (3,2), (4,5)\}$ is a Partial order on set A.

only '1' choice
5
4

How many total order relation are possible on set A such that relation 'R' is included in each total order relation

$$\left.\right\} \; {}^5C_3 * 1 * {}^2C_2 * 1$$

$= 10$ total order

H.W. Q:- Let $A = \{1, 2, 3, 4, 5\}$

Reflexive
Antisymmetric
Transitive

How many partial order relation are possible on set A, such that the Cardinality of each Partial order relation is exactly '7'

Required Relⁿ will be of the form : $\{(1,1), (2,2), (3,3), (4,4), (5,5), \underbrace{(\ ,\ )\ (\ ,\ )}\}$

Two order pairs must be Chosen s.t. Anti-symmetric & Transitive Property is satisfied by relⁿ

Required Relⁿ will be of the form : $\{(1,1),(2,2),(3,3),(4,4),(5,5),(\ ,\ )(\ ,\ )\}$

Two non-diagonal order pairs can be defined using

① Exactly '2' elements {let a & b} i.e. of type $(a,b),(b,a)$ — Not allowed w.r.t anti-symmetric

(or)

② Exactly '3' elements {let a, b & c}

- (a,b) & (b,c) — Not allowed w.r.t. Transitivity
- (a,b) & (a,c)
- (b,a) & (c,a)

} Such order pairs are allowed

(or)

③ Exactly '4' elements {let a, b, c & d} → (a,b) & (c,d) } Any combination w.r.t. '4' elements will be allowed

Required Relⁿ will be of the form $= \{(1,1),(2,2),(3,3),(4,4),(5,5),(\ ,\ )(\ ,\ )\}$

Two non-diagonal order pairs can be defined using

① Exactly '2' elements i.e. of type $(a,b),(b,a)$ ← Not allowed w.r.t anti-symmetric
{let a & b}

(or)

② Exactly '3' elements {let a, b & c}

(a,b) & (b,c) ← Not allowed w.r.t. Transitivity

(a,b) & (a,c)

(b,a) & (c,a)

$\left.\right] {}^5C_3 * (3+3) = 60$

$(a,b),(a,c) = (a,c),(a,b)$
$(b,a),(b,c)$
$(c,a)\ (c,b)$
$(b,a),(c,a)$
$(c,b)\ (a,b)$
$(a,c),(b,c)$

(or)

③ Exactly '4' elements {let a, b, c & d}

(a,b) & (c,d)

(c,d) & (a,b)

$\left.\right] {}^5C_4 * \dfrac{({}^4C_2 * 2) * ({}^2C_2 * 2)}{2!} = 60$

∴ Total no. of partial order on set A of cardinality = 7 are $60 + 60 = 120$

HW Check whether the statement is True or False ?

1. If fog is onto then 'f' must be onto (True)
   If f is not onto then fog can not be onto
2. If 'f' is onto then 'fog' is onto (False)
3. If 'f' is one-one then fog is one-one (False)
4. If 'fog' is one-one then 'f' must be one-one (False)
5. If 'fog' is one-one, then 'g' must be one-one (True)
   if 'g' is not one-one then fog can never be one-one

Q:- $Z_n = \{0, 1, 2, \ldots\ldots, n-1\}$ is a group w.r.t. binary op^n $\oplus_n$.

- In the group $Z_2 \times Z_3$ how many elements are their own inverse.

$$Z_2 = \{0, 1\}$$

$$Z_3 = \{0, 1, 2\}$$

$Z_2 = \{0, 1\}$

$Z_3 = \{0, 1, 2\}$

$Z_2 \times Z_3 = \{(0,0), (0,1), (0,2), (1,0), (1,1), (1,2)\}$

$Z_2 \times Z_3 = \{(a,b) \mid a \in Z_2 \text{ and } b \in Z_3\}$

If $(p,q), (r,s) \in Z_2 \times Z_3$

then $(p,q) * (r,s) = (p \oplus_2 r, q \oplus_3 s)$

$*$: Binary op^n w.r.t. group $Z_2 \times Z_3$

$\oplus_2$: op^n w.r.t. Z_2

$\oplus_3$: op^n w.r.t. Z_3

identity : $(0, 0)$

$\boxed{(0,0) * (0,0) = (0,0)}$

$(0,1) * (0,1) = (0,2)$

$(0,2) * (0,2) = (0,1)$

$\boxed{(1,0) * (1,0) = (0,0)}$

$(1,1) * (1,1) = (0,2)$

$(1,2) * (1,2) = (0,1)$

<u>Note</u>: let S_1 is a group s.t. 'm' elements of the group are inverse of their own

&

let S_2 is a group s.t. 'n' elements of the group are inverse of their own

<u>Then</u>, $S_1 \times S_2$ is also a group

and "$m \times n$" elements in $S_1 \times S_2$ will be inverse of their own.

Note: let S_1 is a group s.t. 'm' elements of the group are inverse of their own

& let S_2 is a group s.t. 'n' elements of the group are inverse of their own

& let S_3 is a group s.t. 'p' elements of the group are inverse of their own

Then, $S_1 \times S_2 \times S_3$ is also a group

and "$m \times n \times p$" elements in $S_1 \times S_2 \times S_3$ will be inverse of their own.

2 mins Summary

Topic GATE PYQ

Topic Degree sequence

Topic Graphic

Slide

THANK - YOU

Slide